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EDITORIAL

Focus on game theory and AI in complex systems

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Abstract

This editorial reviews recent progress at the intersection of game theory and artificial intelligence (AI) in tackling complex systems related to human behaviour, as featured in the Focus Issue ('on Game Theory and AI on Complex Systems') of *J. Phys. Complex*. Leveraging game-theoretic frameworks alongside AI-driven analysis, contributions in this issue explore a broad range of challenges, from modelling behavioural dynamics to creating incentives for pro-social behaviour in hybrid human-AI societies. The contributions underline the potential of this interdisciplinary approach to advancing our understanding of complex human interactions and address critical societal challenges.

1. Introduction

Human behaviour, shaped by individual choices and collective interactions, underpins the complexities of societal systems. Understanding these complexities requires a multi-faceted approach that integrates diverse perspectives and analytical tools (Perc *et al* 2017). Game theory and its extensions such as evolutionary game theory, provide a powerful lens for analysing strategic decision-making in situations involving multiple interacting agents (Hofbauer and Sigmund 1998). They allows us to model the dynamics of cooperation, competition, and the emergence of collective behaviours (Nowak 2006). However, the complexity of real-world scenarios often surpasses the analytical tractability of traditional game-theoretic models (Han 2022, Lu *et al* 2024, Capraro *et al* 2024a, Hammond *et al* 2025). This is where recent advances in Artificial Intelligence (AI) can prove useful. AI's capacity for processing vast datasets, identifying patterns, and optimising complex systems offers a crucial complement to game-theoretic analysis.

This Focus Issue explores the synergistic potential of game theory and AI in addressing the complexities of human behaviour within complex systems. We bring together latest research on theoretical modelling and AI-driven analysis to investigate diverse aspects of human interaction, from behavioural dynamics and learning processes, dynamics of cooperation hybrid human-AI populations to the emergence of collective phenomena. Empirical behavioural studies further enhance our understanding by grounding theoretical insights in real-world observations. The contributions presented here aim to advance our knowledge of human behaviour, inform the design of effective interventions, and offer new perspectives on addressing significant societal challenges.

2. Behavioural and learning dynamics in social dilemmas

The study of mechanisms that effectively promote the evolution of prosocial behaviours, like cooperation, coordination and climate change mitigation, has gained significant interest. However, a major challenge in mathematical modelling lies in accounting for the complexities of cognitive and emotional capacities and diverse personalities. The papers published in this issue aim to bridge this gap by examining factors such as individual memory sizes, learning abilities, and various cognitive theories.

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Zhang *et al* (2024) investigate the emergence of cooperation in a spatial donation game using a Q-learning approach. The authors show that cooperation emerges in this setting, with the rate of cooperation increasing with a lower learning rate and higher discount factor. Analysis of the Q-table evolution reveals that this cooperation arises from a trade-off between immediate and long-term rewards, highlighting the role of individual learning and environmental feedback.

Cui *et al* (2024) study the evolutionary dynamics of a multiplayer snowdrift game with memory effects on simplicial complexes, where players can choose to cooperate, defect, or exit. The authors show that memory effects in networked populations can inhibit defection and promote cooperation. The interplay between the payoff for exiting, the structure of the simplicial complex, and memory length influences the coexistence of different player types.

Muñoz-Álvarez *et al* (2024) explore the dynamics of natural resource exploitation using an agent-based model incorporating Cognitive Hierarchy Theory. The model examines how agents' assumptions about others' behaviour, combined with resource dynamics and long-term perspectives, affect resource sustainability and individual benefits. The authors show that moderate levels of strategic reasoning and a long-term perspective lead to more sustainable resource use.

Vasellini *et al* (2024) study the emergence of cooperation in a repeated Prisoner's Dilemma game, using agent-based simulations. Instead of relying on traditional image scores, the authors introduce a 'private list' mechanism where agents selectively choose opponents based on their past cooperation and share this information (aka gossip) under certain conditions. They show that private list sharing fosters cooperation and leads to the emergence of central hubs within a spatial population, suggesting that centralised structures can support cooperation.

Harper and Safyan (2024) explore the use of momentum, a concept from machine learning, within evolutionary dynamics, showing that momentum can accelerate the convergence of evolutionary processes, including the continuous and discrete replicator equations and Euclidean gradient descent on populations. The study also shows that momentum can alter the convergence properties of these dynamics, potentially leading to divergence or breaking cyclical patterns in certain scenarios.

Finally, using economic experiments with human participants, Alfonso *et al* (2024) study the challenges of climate change mitigation in a collective-risk social dilemma. The authors reveal a significant gap between the collective action required to avert a hypothetical catastrophe and the actual contributions made, even when the risk is high. The study highlights the complexity of communication and the influence of individual perceptions and risk aversion on collective action in large groups.

3. Large language models (LLMs) and game theory

The advent of LLMs offers to transform research across natural and social sciences, offering new paradigms for understanding complex systems (Lu *et al* 2024). In the context of Game Theory, LLMs offer new opportunities thanks to the possibility of creating open-ended agents. This issue contributes to this active research at the intersection of advance AI and game theory, exploring how LLMs can simulate psychological dispositions and trust-like behaviours, with implications for understanding dynamics of human-AI interactions (Zimmaro *et al* 2024).

In particular, Phelps and Russell (2025) investigate whether GPT models can simulate different psychological dispositions (altruism, cooperation, competitiveness, selfishness) in economic games (Dictator Game and Prisoner's Dilemma). Using prompt engineering, they assigned 'personalities' to GPT-3.5 and observed its behaviour in the games. They show that cooperation levels generally aligned with the assigned personalities, supporting the potential of LLMs for studying behavioural economics.

On the other hand, Johnson and Obradovich (2024) study whether an AI language model (in particular, OpenAI's text-davinci-003) exhibits trust-like behaviour toward humans using a trust game with real monetary incentives. They show that the AI model demonstrates trust more frequently when real incentives are involved compared to hypothetical scenarios, suggesting that AI's trust-related behaviour is influenced by actual consequences. These findings are robust to various experimental manipulations.

4. Game theory of incentives and interventions

Designing effective incentives and interventions for positive behavioural change in dynamic multi-agent systems has been a core issue in game theory and AI (Phelps *et al* 2010, Balliet *et al* 2011). Contributing to this direction, works in this issue examine innovative reward mechanisms to foster cooperation in social dilemmas, the interplay of network structure and incentives on cooperative behaviour, strategies for

controlling infectious diseases through social norm adherence, and effective incentive models to enhance the safety of AI development.

Xiao *et al* (2023) study a 'sampling reward' mechanism for promoting the evolution of cooperation in public goods games and collective-risk social dilemmas. In involves rewarding cooperators based on the proportion of cooperators in a randomly sampled subset of the population, offering a cost-effective alternative to rewarding all cooperators. They show that high reward thresholds, small sample sizes, and high reward intensities promote cooperation, but that increasing reward intensity beyond a certain point yields diminishing returns.

Jiang *et al* (2025) investigate the combined effects of incentive mechanisms and network coupling on cooperation in a two-layer network model of multigames (Prisoner's Dilemma and Public Goods Game). They show that a combination of moderate coupling strength and an incentive scheme that redistributes wealth from defectors to low-income cooperators leads to higher cooperation rates and lower inequality, particularly in regular networks. The results highlight the dynamic interplay between network structure, incentives, and wealth distribution in shaping cooperative behaviour.

Vilone *et al* (2024) explore how varying the level of adherence to social norms impact the spread of infectious diseases, using a modified susceptible–infected–recovered model. They show that heterogeneity in compliance of the preventive measures, such as hygiene and social distancing, can make a population more resilient to epidemics. The research provide important insights for designing effective interventions, where focusing on achieving full compliance within specific subgroups may be more effective than aiming for average compliance across the entire population.

Bova *et al* (2024) explore how to incentivise AI auditors to enhance the safe development and deployment of AI systems. They compare 'Adversarial Incentives,' which reward auditors only for catching unsafe behaviour, and 'Vigilant Incentives,' which reward auditors consistently unless they fail to detect unsafe behaviour, showing the latter is more effective in most cases. Evolutionary game theory was used to model these scenarios and provide insights for designing effective incentive strategies for a robust AI auditing ecosystem.

5. Conclusion and directions

This Focus Issue contributes to the advancements made by combining Game Theory and AI to address complex systems problems related to human behaviours. The diverse contributions presented highlight the strength of this interdisciplinary approach, providing novel perspectives on behavioural and learning dynamics, the design of effective incentive and intervention mechanisms in diverse domains, and the capacity of LLMs to replicate human behaviour.

We now comment on some directions for future research in this fascinating area. First, as AI agents are increasingly more prevalent in our society, further investigation into dynamics of hybrid human-AI systems is essential to understand the complexity of trust, cooperation, and competition in this hybrid setting, which is vital for ethical and responsible AI deployment (Guo *et al* 2023, Santos 2024). Second, leveraging LLMs for policy design offers opportunities to refine interventions before real-world application, potentially leading to more effective and equitable solutions (Buscemi *et al* 2025). Third, using advance machine learning approaches for developing efficient incentive and intervention mechanisms in large populations and complex networks, including data-driven personalised incentives, is another crucial direction. Finally, addressing the ethical implications of AI integration into social systems, with a focus on fairness, transparency, and accountability, is critical (Capraro *et al* 2024b). By focusing on these areas, we can fully realise the potential of game theory and AI to build a more cooperative, sustainable, and equitable society.

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